

11.8 Power Series

1 Definition

A **Power Series** is essentially a polynomial with infinite degree.

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

The domain of a power series is the set of x -values for which it converges.

The domain is also called the **Interval of Convergence (IOC)**.

Finding the IOC of a power series

1. use the ratio test to get L .
2. set $|L| < 1$ and solve for x . This gives the endpoints of the IOC.
3. Plug each endpoint into x in the series and determine if it converges or diverges. If it converges, include the endpoint in the IOC.

2 Examples

For each example, determine the interval of convergence for the given series.

2.0.1 Example:

$$\sum_{n=1}^{\infty} x^n \quad \lim_{n \rightarrow \infty} \frac{x^{n+1}}{x^n} = \lim_{n \rightarrow \infty} \frac{\cancel{x^n} \cdot x}{\cancel{x^n}} = x \quad -1 < x < 1$$

I.O.C.

check endpoints (plug in for x)

$$\underline{x = -1}$$

$$\sum (-1)^n = -1 + 1 - 1 + 1 - \dots \quad \mathcal{D} \leftarrow \text{Diverges, so don't include in IOC}$$

$$\underline{x = 1}$$

$$\sum 1^n = 1 + 1 + 1 + \dots \quad \mathcal{D}$$

$$\boxed{(-1, 1)}$$

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{x^n}{n} \quad \lim_{n \rightarrow \infty} \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} = \lim_{n \rightarrow \infty} \frac{\cancel{x^n} \cdot x}{\cancel{x^n}} \cdot \frac{\cancel{n}}{n+1} = x$$

$$-1 < x < 1$$

$$x = -1$$

$$\sum \frac{(-1)^n}{n} \text{ C. (A.S.T.) } \leftarrow \text{include } -1$$

$$\boxed{\text{I.O.C.} \\ [-1, 1)}$$

$$x = 1$$

$$\sum \frac{1}{n} \text{ D}$$

2.0.3 Example:

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{2^n} \quad \lim_{n \rightarrow \infty} \frac{(x-1)^{n+1}}{2^{n+1}} \cdot \frac{2^n}{(x-1)^n} = \lim_{n \rightarrow \infty} \frac{\cancel{(x-1)^n} \cdot (x-1)}{\cancel{(x-1)^n}} \cdot \frac{\cancel{2^n}}{\cancel{2^n} \cdot 2}$$

$$= \frac{x-1}{2}$$

$$-1 < \frac{x-1}{2} < 1 \Rightarrow -2 < x-1 < 2$$

$$-1 < x < 3$$

$$x = -1$$

$$\sum \frac{(-1-1)^n}{2^n} = \sum \frac{(-2)^n}{2^n} = \sum \frac{(-1)^n \cdot \cancel{2^n}}{\cancel{2^n}} = \sum (-1)^n \text{ D}$$

$$\boxed{\text{I.O.C.} \\ (-1, 3)}$$

$$x = 3$$

$$\sum \frac{(3-1)^n}{2^n} = \sum \frac{2^n}{2^n} = \sum 1 \text{ D}$$

2.0.4 Example:

$$\sum_{n=1}^{\infty} \frac{n^x}{n!} \quad \lim_{n \rightarrow \infty} \frac{(n+1)^x}{(n+1)!} \cdot \frac{n!}{n^x} = \lim_{n \rightarrow \infty} \frac{(n+1)^x}{n^x} \cdot \frac{\cancel{n!}}{(n+1)\cancel{n!}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^x + \dots}{n^x} \cdot \frac{1}{n+1} = 1 \cdot 0 = 0$$

$-1 < 0 < 1 \leftarrow$ For which values of x is this true?
 A: All values of x

$$\text{IOC} \\ (-\infty, \infty)$$

2.0.5 Example:

$$\sum_{n=1}^{\infty} (-3x)^n \quad \lim_{n \rightarrow \infty} \frac{(-3x)^{n+1}}{(-3x)^n} = \lim_{n \rightarrow \infty} \frac{\cancel{(-3x)^n} \cdot 3x}{\cancel{(-3x)^n}} = 3x$$

$$\underline{x = -\frac{1}{3}}$$

$$\sum \left(-3\left(-\frac{1}{3}\right)\right)^n = \sum 1^n \quad \mathcal{D}$$

$$\underline{x = \frac{1}{3}}$$

$$\sum \left(-3\left(\frac{1}{3}\right)\right)^n = \sum (-1)^n \quad \mathcal{D}$$

$$-1 < 3x < 1$$

$$-\frac{1}{3} < x < \frac{1}{3}$$

$$\left(-\frac{1}{3}, \frac{1}{3}\right)$$

2.0.6 Example:

$$\sum_{n=1}^{\infty} \frac{(x+2)^n}{\sqrt{n}} \quad \lim_{n \rightarrow \infty} \frac{(x+2)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{(x+2)^n} = \lim_{n \rightarrow \infty} \frac{\cancel{(x+2)^n} (x+2)}{\cancel{(x+2)^n}} \cdot \frac{\cancel{\sqrt{n}}}{\sqrt{n+1}} \xrightarrow{!}$$

$$= x+2$$

$$-1 < x+2 < 1$$

$$-3 < x < -1$$

$$\boxed{\text{IOC} \\ [-3, -1)}$$

$$\underline{x = -3} \\ \sum \frac{(-3+2)^n}{\sqrt{n}} = \sum \frac{(-1)^n}{\sqrt{n}} \quad \mathbb{C}$$

$$\underline{x = -1} \\ \sum \frac{(-1+2)^n}{\sqrt{n}} = \sum \frac{1}{\sqrt{n}} \quad \mathbb{D}$$

2.0.7 Example:

$$\sum_{n=1}^{\infty} x^n \cdot n! \quad \lim_{n \rightarrow \infty} \frac{x^{n+1} \cdot (n+1)!}{x^n \cdot n!} = \lim_{n \rightarrow \infty} \frac{\cancel{x^n} \cdot x \cdot \cancel{(n+1)!}}{\cancel{x^n} \cdot \cancel{n!}} = \lim_{n \rightarrow \infty} x \cdot (n+1)$$

$$= \lim_{n \rightarrow \infty} x(n+1) = \begin{cases} \infty & \text{if } x > 0 \\ 0 & \text{if } x = 0 \leftarrow x = 0 \text{ is the only value for which it converges.} \\ -\infty & \text{if } x < 0 \end{cases}$$

$$\boxed{\text{"IOC"} \\ \{0\}}$$

Power Series Worksheet 1

1. Find the interval of convergence for ...

$$\sum_{n=0}^{\infty} \frac{(2x+1)^n}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{(2x+1)^{n+1}}{(2x+1)^n} \cdot \frac{2^n}{2^{n+1}} = \boxed{\frac{2x+1}{2}}$$

$$\begin{aligned} -1 &< \frac{2x+1}{2} < 1 \\ -2 &< 2x+1 < 2 \\ -3 &< 2x < 1 \\ -\frac{3}{2} &< x < \frac{1}{2} \end{aligned}$$

$$\frac{x = -\frac{3}{2}}{\sum \frac{(2(-\frac{3}{2})+1)^n}{2^n} = \sum \frac{(-2)^n}{2^n} = \sum (-1)^n \mathcal{D}}$$

$$\boxed{\left(-\frac{3}{2}, \frac{1}{2}\right)}$$

$$\frac{x = \frac{1}{2}}{\sum \frac{(2(\frac{1}{2})+1)^n}{2^n} = \sum 1 \mathcal{D}}$$

2. $\sum_{n=1}^{\infty} \frac{(x-1)^n}{n}$

$$\lim_{n \rightarrow \infty} \frac{(x-1)^{n+1}}{n+1} \cdot \frac{n}{(x-1)^n} = \lim_{n \rightarrow \infty} \frac{(x-1)^{n+1}}{(x-1)^n} \cdot \frac{n}{n+1}$$

$$= x-1 \quad \begin{aligned} -1 &< x-1 < 1 \\ 0 &< x < 2 \end{aligned}$$

$$\frac{x=0}{\sum \frac{(-1)^n}{n} \mathcal{C}} \quad \frac{x=2}{\sum \frac{1}{n} \mathcal{D}}$$

$$\boxed{[0, 2)}$$

$$3. \sum_{n=0}^{\infty} nx^n \quad \lim_{n \rightarrow \infty} \frac{(n+1)x^{n+1}}{n x^n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{\cancel{x^n} \cdot x}{\cancel{x^n}} = x$$

$$-1 < x < 1$$

$$\boxed{(-1, 1)}$$

$$\underline{x = -1}$$

$$\sum n (-1)^n \mathbb{D}$$

$$\underline{x = 1}$$

$$\sum n \mathbb{D}$$

$$4. \sum_{n=1}^{\infty} \frac{x^n}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} = \lim_{n \rightarrow \infty} \frac{\cancel{n!}}{(n+1)\cancel{n!}} \cdot \frac{x^{n+1}}{x^n}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n+1} \cdot \frac{\cancel{x^n} \cdot x}{\cancel{x^n}} = 0$$

$$\begin{array}{c} -1 < 0 < 1 \\ \boxed{(-\infty, \infty)} \end{array}$$

$$1. \left(-\frac{3}{2}, \frac{1}{2}\right)$$

$$2. [0, 2)$$

$$3. (-1, 1)$$

$$4. (-\infty, \infty)$$